

SIADS 699 Capstone Project Report

Team 27 - Predicting Film Financial Success

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Introduction

Film success is inherently uncertain, with outcomes influenced by a combination of factors such as budget, genre, cast, and creative direction. While individual movies can be unpredictable, broader patterns may emerge when analyzing financial performance, audience reception, and production scale at an aggregate level.

Our project investigates whether movie success can be predicted using pre-release characteristics. Specifically, we aim to determine how features such as genre, production budget, and actor and director “star power” relate to the likelihood that a film achieves financial success.

We started by conducting exploratory data analysis (EDA) to identify relationships between key variables, including how genres, actors, and directors influence financial performance and audience reception over time. Second, we developed a classification model to predict whether a film is financially successful based on pre-release features.

To support this analysis, we constructed a merged dataset from IMDb and TMDb, combining movie metadata, ratings, and financial information. We then engineered features capturing historical performance of actors, directors, and genres, and applied clustering techniques to identify actor and movie archetypes, some of which are incorporated as additional model features.

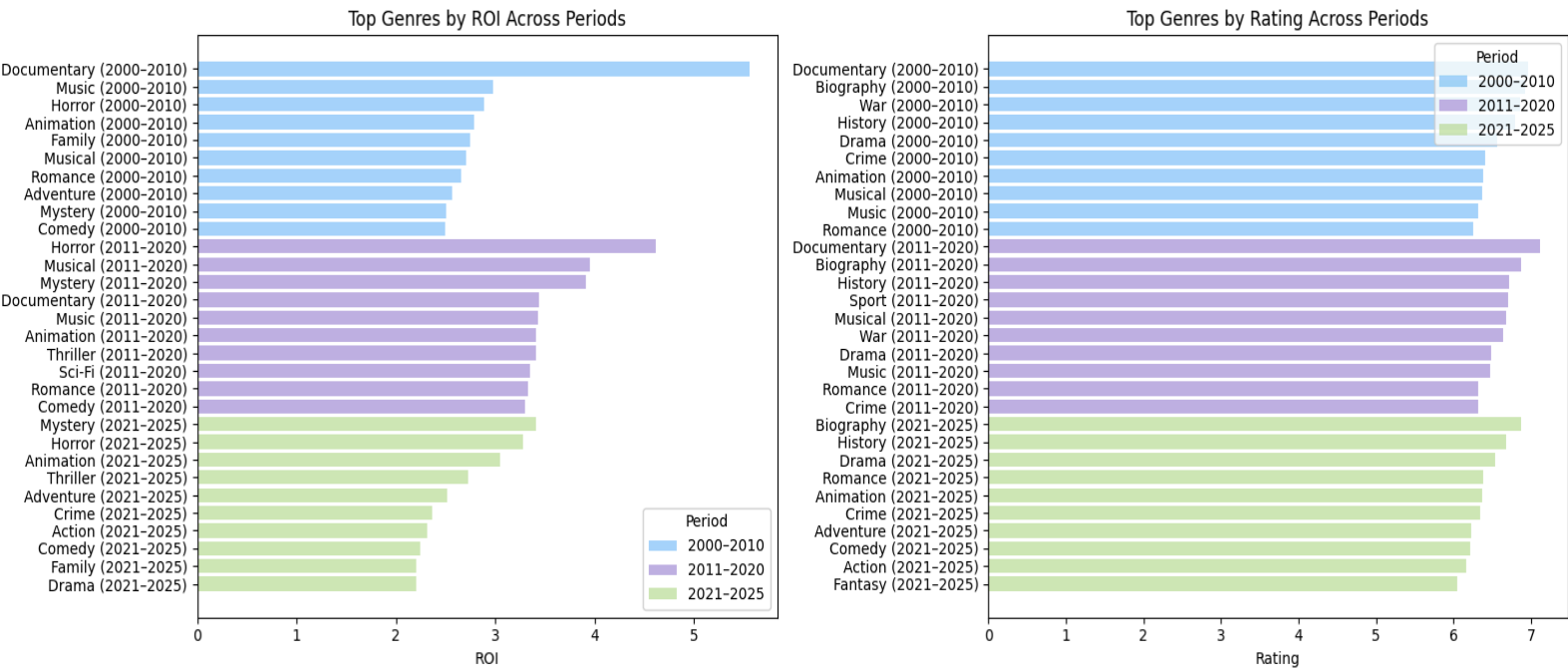
Our findings indicate that movie success is multi-dimensional. Financial efficiency, production scale, and audience reception are driven by distinct factors and are not fully aligned. While large budgets and franchise participation increase scale and visibility, financial success is often achieved through cost-efficient production models, and audience reception is more closely tied to narrative-driven content.

Exploratory Data Analysis

The analysis focuses on three key dimensions of movie success: financial performance (Return on Investment, ROI), audience reception (IMDb ratings), and production scale (budget). The goal is to identify structural relationships between these dimensions and understand how factors such as genre, actors, and directors influence outcomes, both overall and over time. These insights are subsequently used to inform feature engineering decisions and modeling design.

Genre-Level Analysis

Analyzing genres reveals a clear divergence between financial efficiency and audience reception. From a financial perspective, Documentary and Horror films consistently achieve the highest ROI. Documentary films exceed ROI values of 5.5 in earlier periods despite very low budgets (~\$2–6M), while Horror, Mystery, and Musical films reach ROI levels of approximately 3.5–4.6 with modest production costs. These genres follow a low-cost, high-return model where profitability is driven by efficiency rather than scale.



In contrast, high-budget genres such as Animation and Sci-Fi also generate strong returns but with lower efficiency (ROI ~3.3–3.4), indicating diminishing returns at higher production scales. From an audience perspective, Biography, War, and History films consistently achieve the highest ratings (~6.7–6.9), but with lower ROI, suggesting that narrative-driven genres prioritize critical reception over financial efficiency.

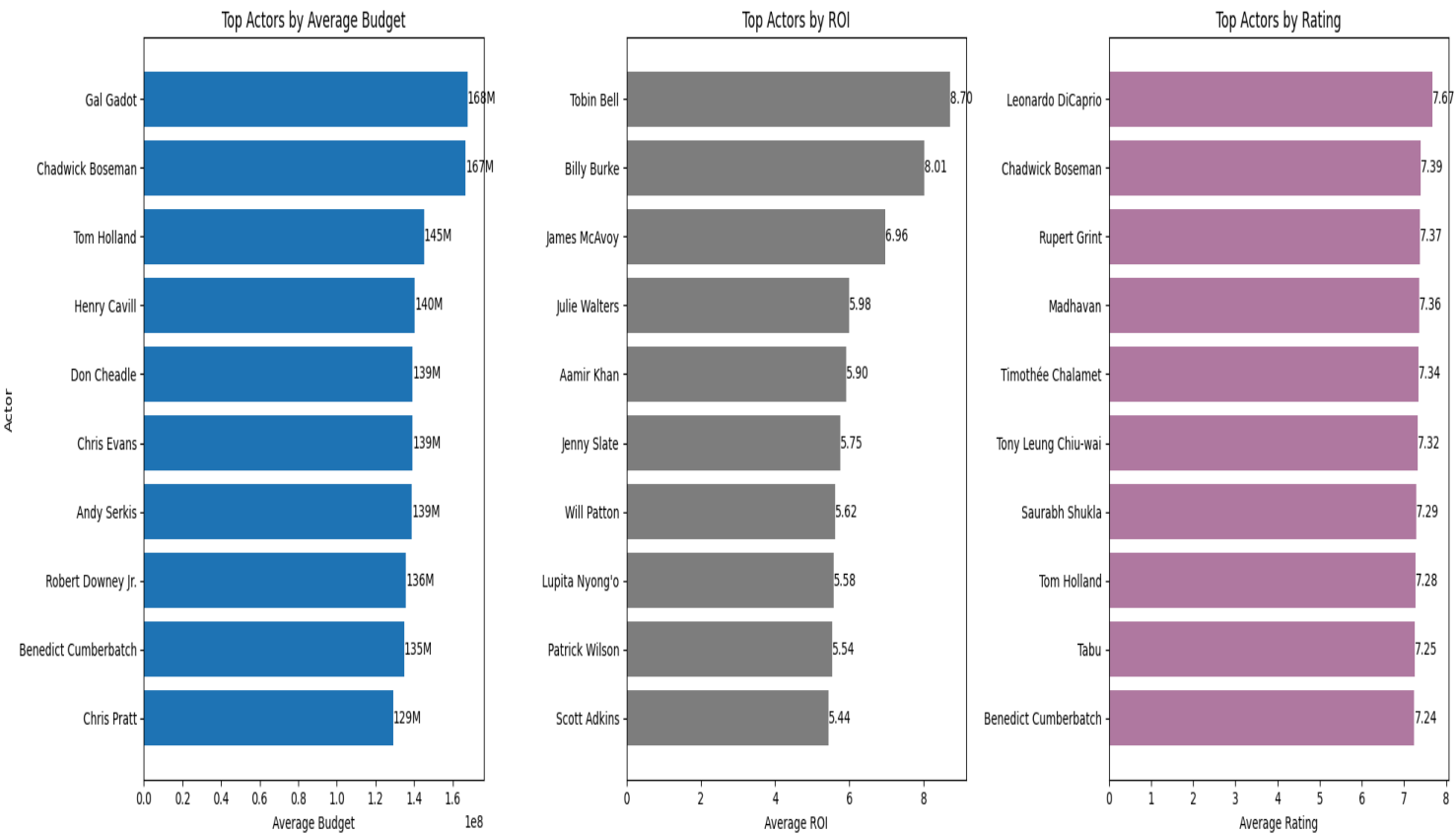
At the genre combination level, financial performance is concentrated within a narrow group, particularly Horror–Mystery–Thriller, which consistently delivers the highest ROI across all periods (peaking around 9.3). In contrast, top-rated combinations remain concentrated in Drama-based categories such as Biography–Drama–History and Crime–Drama, with minimal variation over time.

Examining trends over time shows that financial leadership shifts moderately—from Documentary in early periods to Horror and Mystery in later years—but remains within cost-efficient genres. In contrast, rating leaders remain highly stable, indicating that audience preferences for narrative-driven content do not change significantly over time.

These findings suggest that historical ROI varies significantly across genres and over time, indicating that genre-based features—particularly historical ROI—are likely to carry meaningful predictive signals. However, the divergence between ROI and ratings highlights that financial success is multi-dimensional and cannot be explained by genre alone. Additionally, the observed temporal shifts support the use of a time-based train/validation/test split, as relationships between features and outcomes evolve over time.

Actor-Level Analysis

Initial results show that actors with the highest ROI are often less recognized and associated with lower-budget films. This indicates that high ROI is primarily driven by cost-efficient production models rather than star power. To ensure meaningful comparisons, the analysis is restricted to actors with sufficient experience and audience exposure.



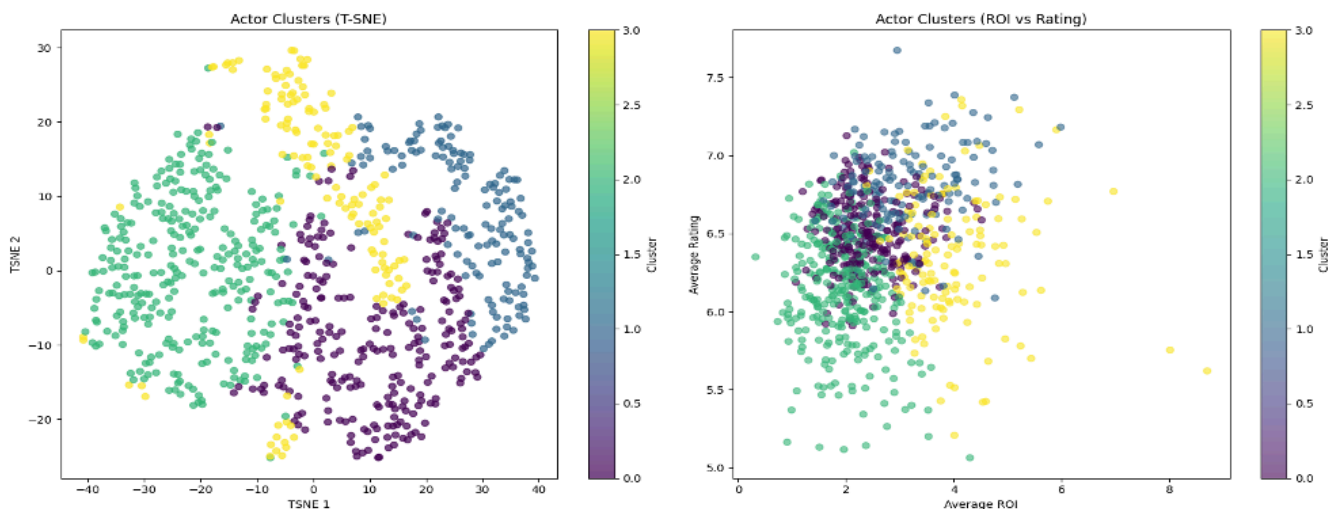
Within this filtered group, three distinct patterns emerge. High-budget actors (e.g., Gal Gadot, Tom Holland) are strongly associated with large-scale franchise productions (~\$130M–\$170M budgets), achieving stable but moderate ROI (~2.5–4.5) and consistent ratings. High-ROI actors (e.g., Tobin Bell) operate in lower to mid-budget films and achieve strong financial efficiency (ROI ~8+), often in genre-driven productions such as horror. High-rating actors (e.g., Timothée Chalamet) deliver strong audience reception (>7.2) with moderate ROI, reflecting a focus on critically successful films. Elberse investigates in a 2007 study on *The Role of Star Power in Movie Success* whether well-known actors significantly influence box office outcomes. The findings suggest that star power alone does not guarantee financial success, emphasizing that broader production and market factors play a critical role. This insight aligns with our analysis of actor and director “star power,” where we observe that financial efficiency, production scale, and audience reception are driven by different underlying factors.

Over time, actor performance evolves from strong alignment in the early 2000s—driven by a small number of dominant franchises—to increasing separation across dimensions. In recent years, a clear structure emerges: budget is driven by franchise actors, ROI by cost-efficient productions, and ratings by critically strong or hybrid mainstream actors. This reflects increasing specialization of actor roles within the industry.

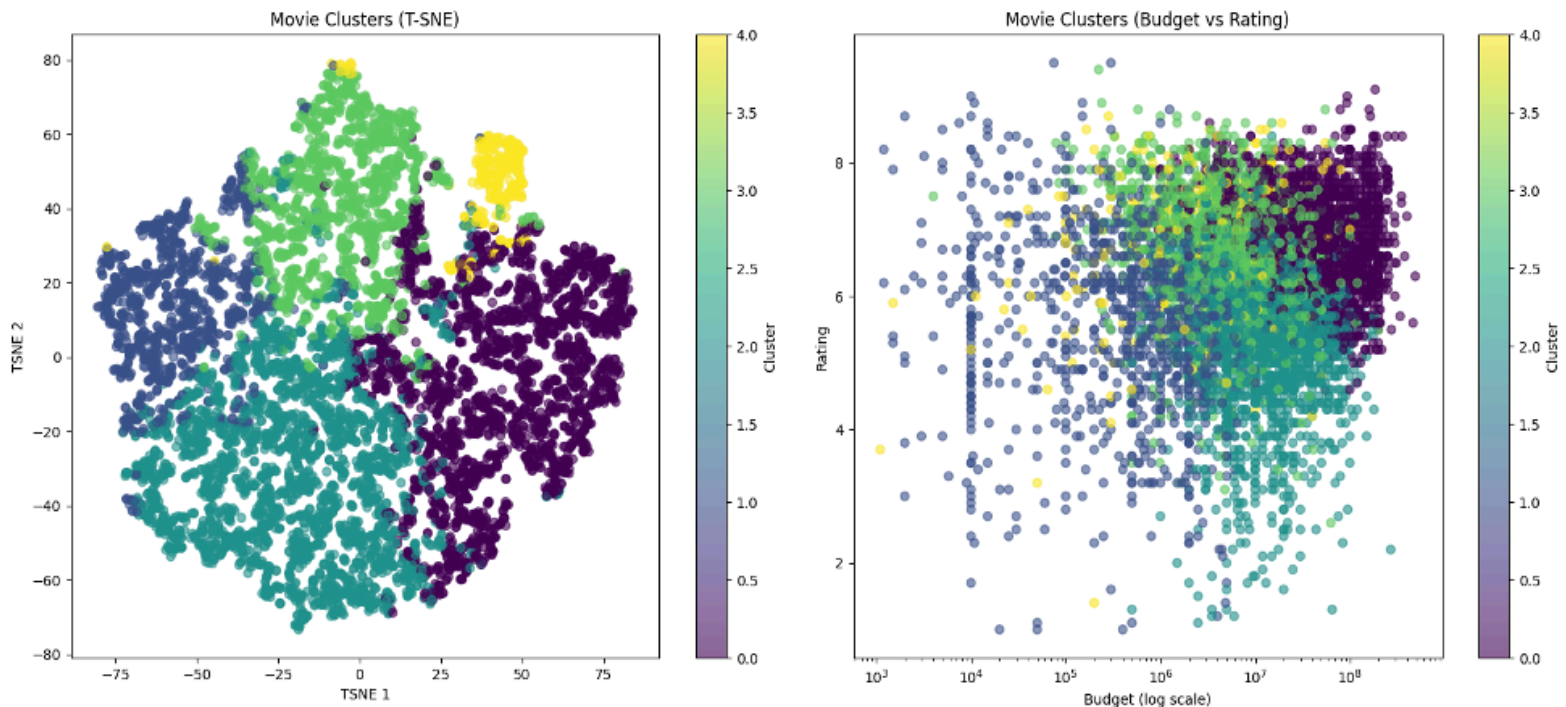
These results highlight that actor performance cannot be captured by simple features or sparse combinations such as actor–genre pairs. Under a temporal split, many such combinations are not observed in the training data, requiring imputation and leading to unstable model performance. Instead, aggregated “star power” features based on historical averages (e.g., ROI, rating, budget, experience, popularity) provide a more robust representation of actor influence. This approach reduces sparsity, avoids excessive imputation, and improves generalization.

Clustering Analysis: Actor and Movie Archetypes

To capture the multi-dimensional nature of performance, we apply K-Means clustering to both actors and movies using features such as ROI, ratings, budget, and popularity. The number of clusters is selected using the Elbow Method and Silhouette Score, balancing statistical fit with interpretability.



Clustering reveals that both actor “star power” (above) and movie performance (below) consist of multiple independent dimensions. For actors, scale (budget and exposure), audience perception (ratings and popularity), and ROI form distinct archetypes. For movies, performance reflects trade-offs between budget, ROI, ratings, and popularity, confirming that high budgets do not guarantee success and the most efficient films are often not the largest.



Clustering provides a structured way to summarize complex interactions into a smaller set of interpretable features. Cluster assignments serve as higher-level representations that reduce dimensionality and avoid sparsity issues, making them suitable for inclusion in classification models.

EDA Synthesis

Across genres, actors, and directors, the analysis shows that financial success is driven by a combination of production scale, efficiency, and audience reception, rather than any single factor. ROI patterns vary across segments and over time, while budget remains a dominant structural driver.

These findings directly informed our modeling approach. We prioritized historical ROI-based features, avoided sparse interaction terms (such as actor–genre combinations), and constructed aggregated features such as star power and clustering-based representations. Additionally, observed temporal shifts in performance justified the use of a time-based train/validation/test split to better reflect real-world prediction settings.

Overall, the analysis highlights that movie success is inherently multi-dimensional and only partially predictable using structured pre-release data, motivating the use of predictive models to quantify these relationships.

Feature Engineering and Model Development

Data Structure, Problem Framing, and Initial Design Decisions

The objective of our analysis was to predict film financial success using *only* pre-release features. We defined success as any movie having revenue $\geq 1.25 \times$ budget, meaning that the movie had at least a 25% profit margin. Though we had data for movies from 2000-2025, we decided to only include data from 2000-2019 due to some inconsistencies in post-covid data observed during our EDA.

During and after covid, fewer movies were being released only in theaters, and more were being released directly on streaming sites like Netflix, HBO, Prime Video, etc. In the IMDb dataset, revenue only measures box office sales which caused many very “successful” streaming platform released movies to show basically no revenue while still having massive budgets. With no flag to filter out these movies and thousands of movies with potentially incorrect revenues, we decided to simply remove these years from our model.

We implemented a time-based split, training on films released between 2000–2012, validating on 2013–2015, and testing on 2016–2019. With the temporal nature of our data, this decision was made to eliminate data leakage that could otherwise be present.

The final dataset contained 4,142 training observations, 1,142 validation observations, and 1,484 test observations, with 28 total features. These features included both raw variables, such as budget, movie length in minutes, genre, and more complex historical aggregates capturing prior performance trends of actors and genres. A preliminary examination of the data revealed significant differences across budget levels. In particular, high-budget films (\$100M+) showed extremely high baseline success rates, exceeding 85% across all splits, while lower-budget films showed success rates of 40-60%. This imbalance foreshadowed many of the modeling challenges we encountered.

This model and analysis could be used by studios, individuals, or any company that is interested in financing a movie. Since this model uses pre-release features only, those stakeholders would be able to get an idea of whether or not the movie they are deciding to invest in would be a wise financial decision.

Historical Feature Engineering and Leakage Prevention

A large part of the modeling process was the construction of historical performance features. Based on movie distributions explored in our EDA, we first defined specific budget buckets for our movies, those being \$0-5M, \$5-25M, \$25-100M and \$100M+. For each of the features we engineered, we limited the subset of historical train data to the specific budget bucket any individual movie fell into. This helped to reduce imputing values significantly too high for low budget movies and significantly too low for high budget movies.

The features we engineered included median revenue, ratings, and budgets for specific genres and actors. The goal was to identify genre/actor-specific trends using only information that would have been available prior to a film's release. Since the clustering performed in our exploratory data analysis was done to simply visualize similarities between actors and movies, we decided not to include these clusters as features in our model since the features used to generate those clusters were already present. One relevant study is *An Empirical Analysis of Determinants Driving Hollywood Box Office Success* (Wang, 2025), which examines how factors such as production budget, director characteristics, genre, release timing, and audience ratings influence box office revenue. Using multiple linear regression, the study finds that financial investment, directorial reputation, and genre preferences are key drivers of success. This work directly supports the inclusion of similar features in our model and provides a benchmark for evaluating feature importance.

To ensure the integrity of these features, we created them using only data from the training set. This eliminated leakage and prevented the model from incorporating future information in its predictions. In cases where the actor or genre was never seen in the training set, we implemented a hierarchical imputation strategy, first imputing a group-level median if it was observed, and a global median if it wasn't. Median was selected over mean due to some massive ROI and budget outliers which would've caused overall inflation of these features' values (seen in the table below).

Metric	ROI	Average Rating	Budget
Mean	1,037	6.23	\$26,414,884
Median	1.43	6.40	\$10,000,000
Max	3,000,003	9.50	\$489,900,000
Standard Deviation	52,322	1.15	\$42,394,842

Early iterations of our model that used mean-imputed feature values produced unrealistically strong results in some budget buckets and unrealistically weak results in others. After re-defining our imputation strategy, performance metrics stabilized giving us confidence that the model was capturing genuine signal.

Using historical data to engineer features to predict future performance raised an ethical concern we had to consider – If an actor had poor ratings/ROI historically (in the train set), but had since been involved in more successful, higher rated movies, the model might negatively predict the outcome of their future movies based on those historical performance features, which could hypothetically decrease their likelihood of being booked into a role in a movie. In the real world, this issue could be solved by consistently updating the dates used in the train set to more accurately depict recent historical performance.

Model Selection and Tradeoffs

We evaluated four primary classification models: Decision Tree, Random Forest, Logistic Regression, and Gradient Boosting Classifier. Each model was trained on the same feature set to understand the performance of each model with everything else held constant. Hossein & Miller’s 2018 paper on *Predicting Motion Picture Box Office Performance Using Temporal Tweet Patterns* applies machine learning techniques to classify films as successful or unsuccessful based on social media activity. The authors show that patterns in tweet volume and sentiment differ significantly between high and low performing films. While our project focuses on structured data from IMDb and TMDb, this study demonstrates the viability of a predictive approach to this classification problem.

Model	Accuracy	Precision	Recall	F1
Gradient Boosting	0.624	0.677	0.705	0.691
Random Forest	0.613	0.684	0.649	0.666
Decision Tree	0.606	0.684	0.626	0.654
Logistic Regression	0.619	0.717	0.594	0.649

In the table above, we can see that ensemble methods, particularly Gradient Boosting, delivered the strongest performance. On the validation set, the Gradient Boosting model achieved an accuracy of 0.624, with precision, recall, and F1 scores of 0.677, 0.705, and 0.691, respectively. This performance reflected its ability to model complex interactions between features such as budget, historical actor performance, and genre effects. Random Forest

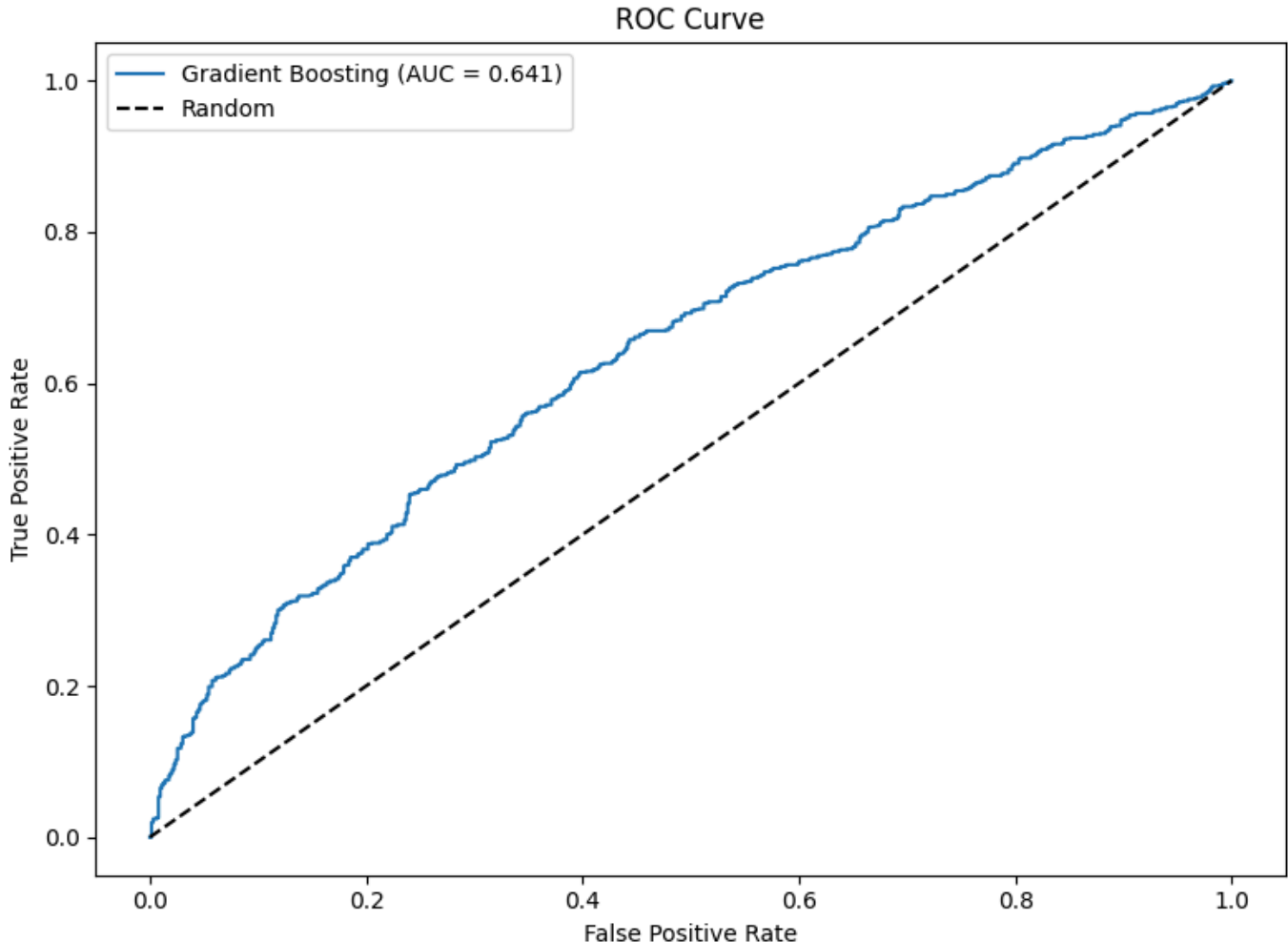
produced slightly weaker results, while the Decision Tree and Logistic Regression performed the worst.

We conducted hyperparameter tuning using the validation set with the goal of balancing model complexity against generalization of performance. While this introduces some risk of overfitting to the validation data, we mitigated this by maintaining a held-out test set for our final model evaluation. We selected the Gradient Boosting Classifier which achieved a test accuracy of 0.613 and an F1 score of 0.674, showing a minor drop from validation performance.

Model Evaluation and Interpretability

ROC-AUC Performance and Model Behavior

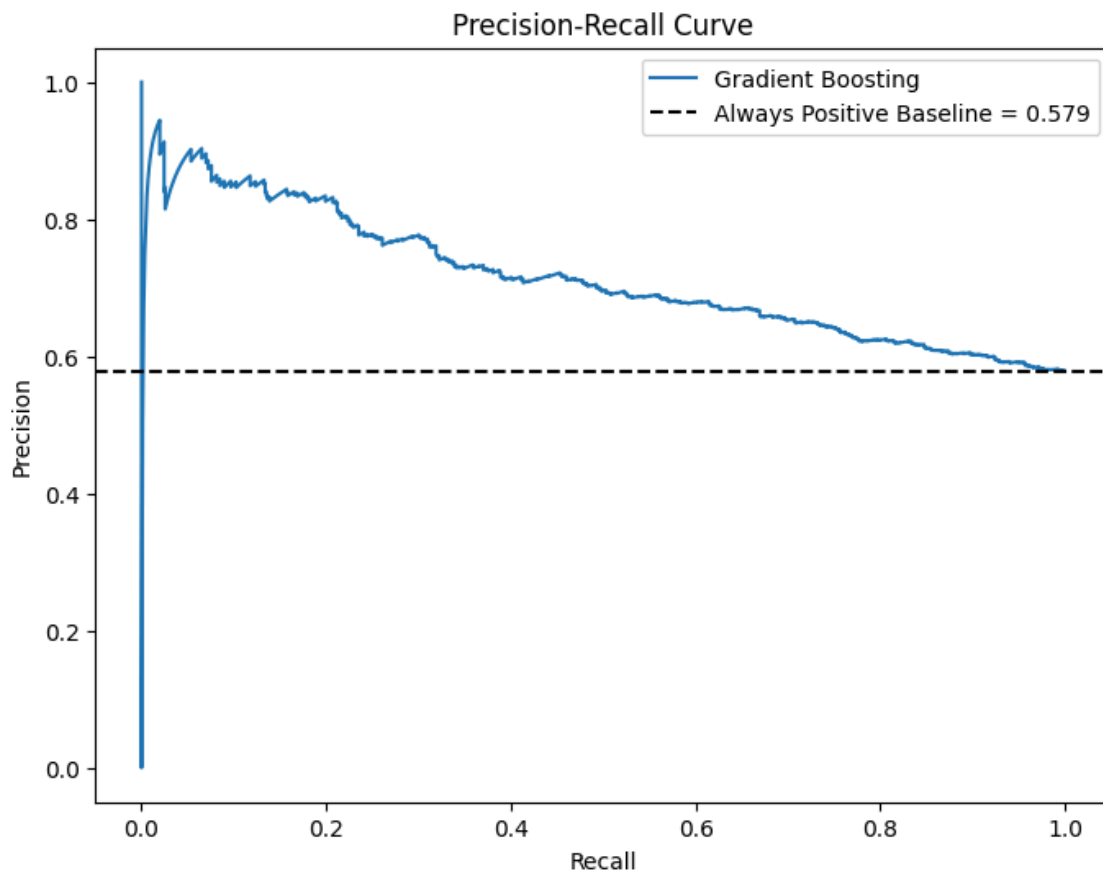
The primary performance metric used for the final model was the area under the ROC curve (AUC). On the test set, the Gradient Boosting classifier achieved an AUC of 0.641, representing a 14.1% improvement over a random classifier and a 7.9% improvement over an all-positive classifier. This confirms that the model is learning patterns from the data, although the magnitude of improvement is small.



Across all models evaluated on the validation set, AUC values were tightly clustered between 0.62 and 0.66, meaning the model architecture was not the primary limiting factor. Instead, predictive performance appears to be limited by feature signal and the inherent variance of film success. While ROC-AUC provides a useful high-level summary, it does not fully capture the effectiveness of the model, particularly in the presence of major class imbalance seen in specific budget buckets.

Precision-Recall Dynamics and Class Imbalance

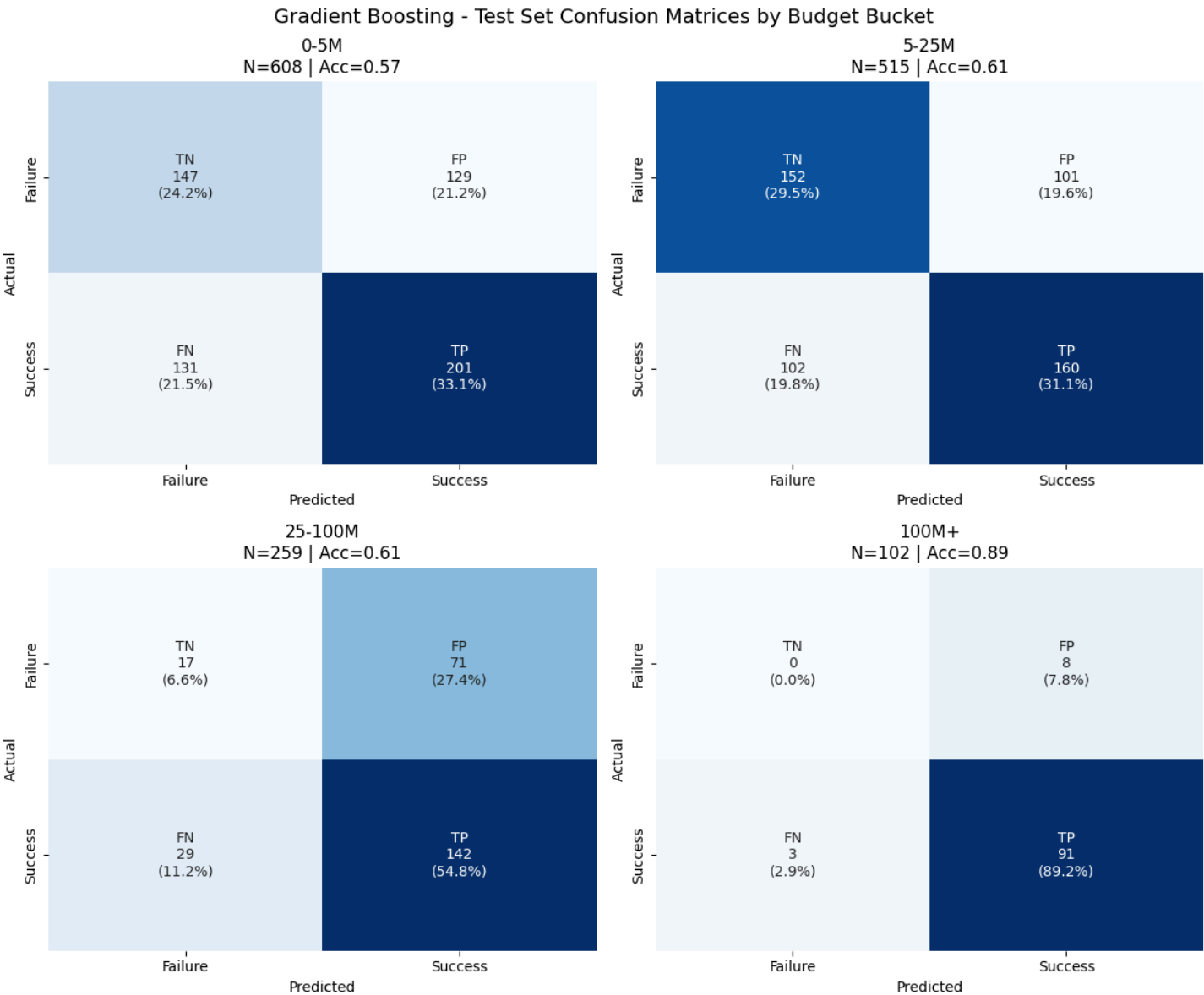
To better understand model performance, we examined precision-recall curves. Precision dropped sharply at low recall thresholds, indicating that the model produces a large number of false positives when attempting to identify successful films.



This behavior is consistent with the underlying class distribution. Since a large proportion of films, especially those with high budgets, are successful, the model is incentivized to predict the positive class more frequently. While this increases recall, it simultaneously decreases precision. From this, we can deduce that the model is effective at capturing a large share of successful films but struggles to capture those high budget movies that end up failing.

Budget-Specific Performance

We conducted a more detailed evaluation by segmenting movies into their respective budget buckets, defined earlier. This analysis revealed that performance varied widely due to the differing level of signal provided by features and major class imbalance seen in each of these segments. In the highest budget segment (\$100M+), the model achieved extremely high precision (0.919) and recall (0.968) due to the very high rate (~85%) of successful movies seen in the train set. However, the corresponding AUC was only 0.535 meaning that for this budget bucket, the model was only marginally better than an all-positive classifier.

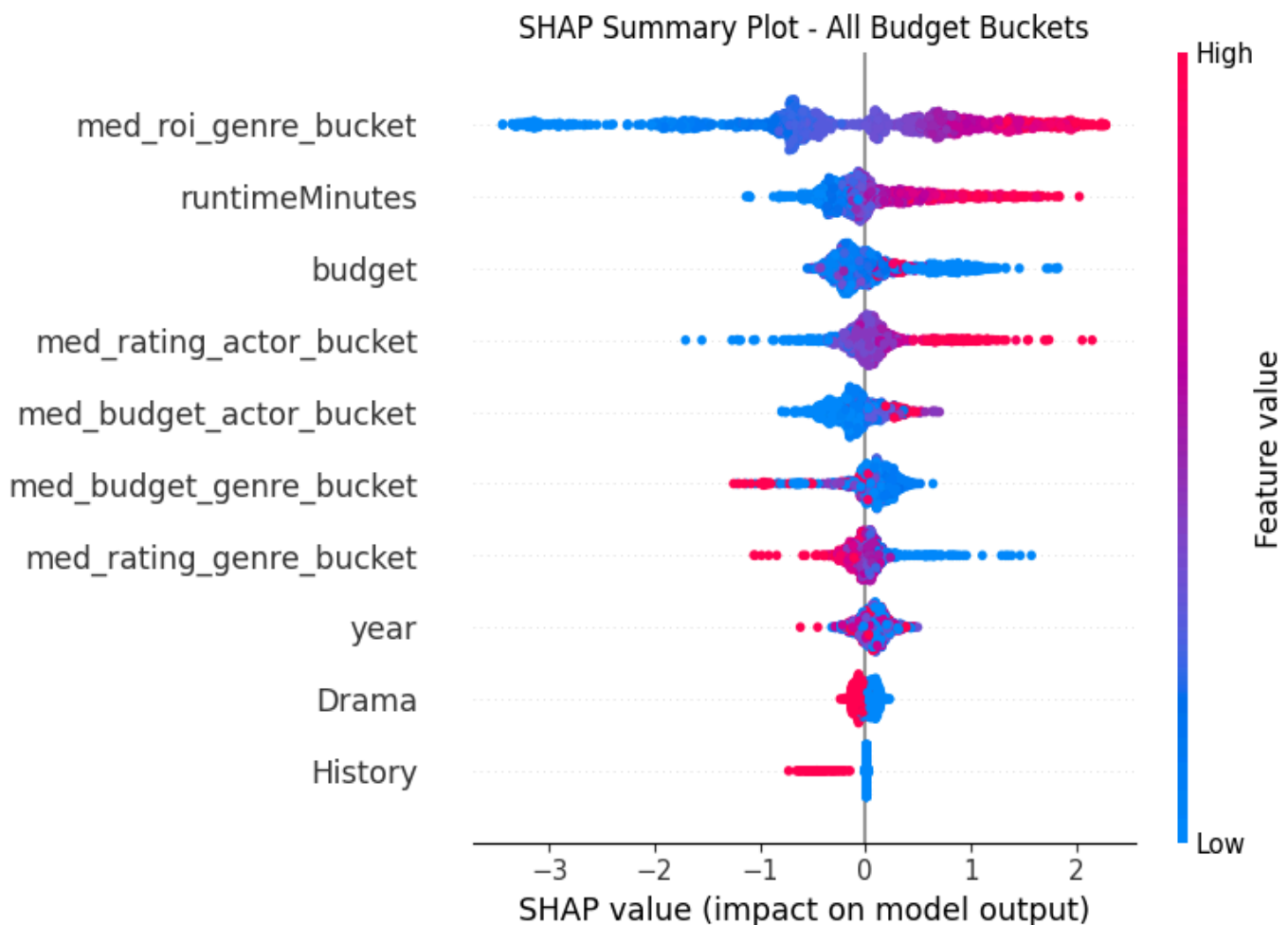


This happens because the model is effectively learning that high-budget films are almost always successful. We can see these results in the budget-bucket specific confusion matrices above which show relatively balanced predictions for the lower budget buckets which transition to majority positive predictions as the budget bucket increases. As a result, the model's predictions within these high budget buckets lack any real separation.

In lower budget categories, performance is more balanced. Precision and recall values are closer to 0.60–0.67, with AUC values around 0.60. These results suggest that while the model captures some signal, it struggles to generalize across different budget buckets.

SHAP Analysis and Feature Contributions

To further interpret our model's performance, we conducted SHapley Additive exPlanations (SHAP) analysis on the Gradient Boosting Classifier using the test set. The results provide insight into which features are driving predictions and how they influence outcomes (top 10 features shown in visualization).



The most influential predictor across budget buckets was the historical median ROI. High values for this feature generally push predictions toward success, meaning that historical ROI for a specific genre is a strong indicator of successful prediction for a film sharing that same genre. Historical performance features, such as average budget and ratings for actors and genres, were also among the top contributors. These features capture patterns related to “star power” and genre trends, and they provide incremental predictive value beyond budget alone.

The SHAP analysis also revealed limited evidence of complex feature interactions. While the Gradient Boosting Classifier is capable of modeling nonlinearity, the dominance of a few key features suggests that the model’s decision-making process is relatively concentrated. This helps explain why performance improvements only slightly improved with hyperparameter tuning.

Overall Synthesis

These results highlight both the strengths and limitations of our modeling approach. The model successfully captures broad patterns of film success and achieves measurable improvements over baseline performance. Conversely, much of this performance is driven by pre-defined signals, particularly budget and historical actor/genre performance, rather than multiple feature interactions.

The combination of ROC-AUC, precision-recall analysis, and SHAP results tell us that while the model is directionally correct, it struggles to make highly selective predictions. In particular, its inability to differentiate within higher budget films shows a key limitation.

This analysis demonstrates that film success is only partially predictable using pre-release data. There are features that we did not have access to, like specific budget breakdown for marketing, actors, and production, film festival inclusion, number of theaters any given movie was released in, actor social media presence, and many other pre-release features that might’ve helped improve performance across budget buckets. While historical performance features provide meaningful signals, they are insufficient to fully capture the complex and often unpredictable factors that drive financial outcomes.

Statement of Work

Evgeny Kuzmin was responsible for data acquisition, preprocessing, and dataset construction, including integration of IMDb and TMDb data, data cleaning, filtering, and preparation of the final modeling dataset. He also conducted EDA to uncover trends across genres, actors, and directors, as well as temporal analysis of these features.

Peter Sklamberg led the model development and evaluation, including feature engineering, implementation of classification models (Logistic Regression, Decision Tree, Random Forest, Gradient Boosting), hyperparameter tuning, and performance evaluation using validation and test sets.

Benjamin Wilson contributed to feature engineering and modeling, including the design and implementation of historical performance features (actor and genre-based), leakage prevention strategies, and support in model evaluation and interpretation.

References

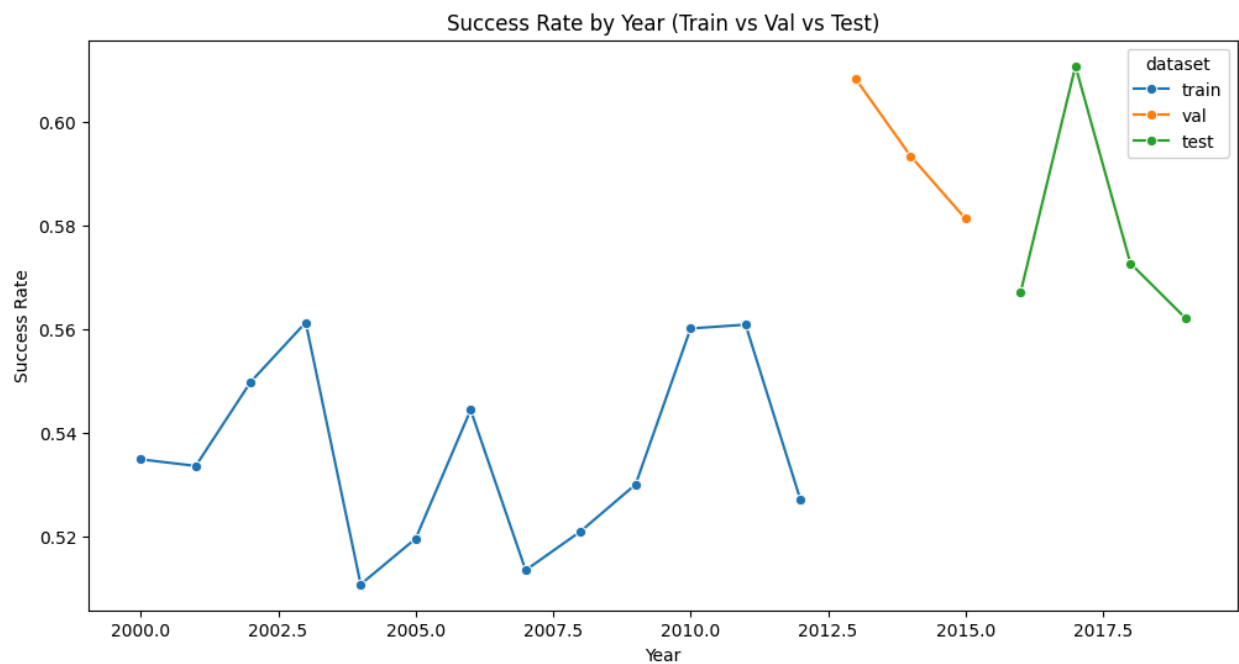
Hosseini, N., and D. W. Miller. "Predicting Motion Picture Box Office Performance Using Temporal Tweet Patterns." *International Journal of Intelligent Computing and Cybernetics*, vol. 11, no. 1, 2018, pp. 64–80.

Elberse, Anita. *Do Star Actors Drive the Success of Movies?* Harvard Business School, 2007.

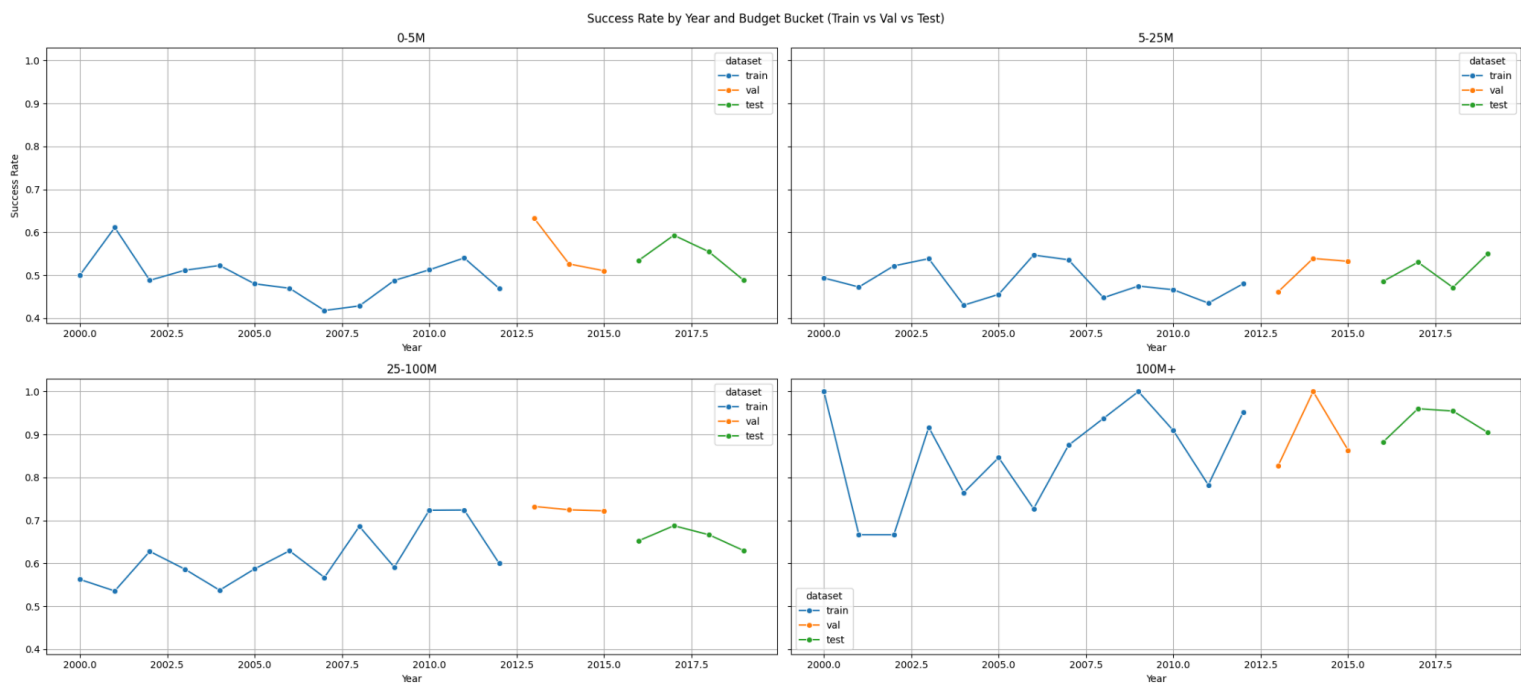
Wang, Siyi. *An Empirical Analysis of Determinants Driving Hollywood Box Office Success*. Proceedings of the 2025 10th International Conference on Financial Innovation and Economic Development (ICFIED 2025), Atlantis Press, 2025.

Appendix

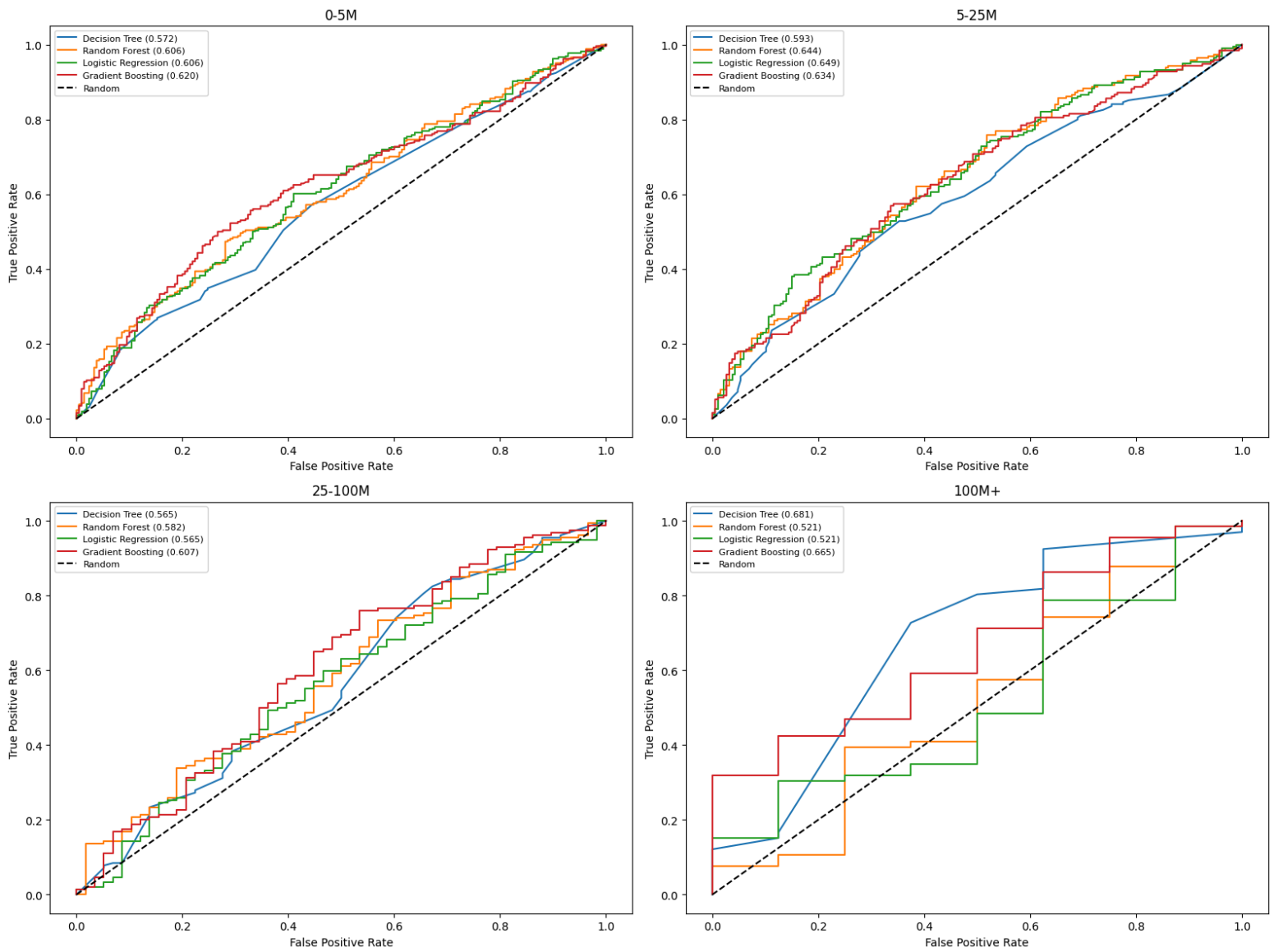
Overall Movie Success Rate broken out by Year and Dataset



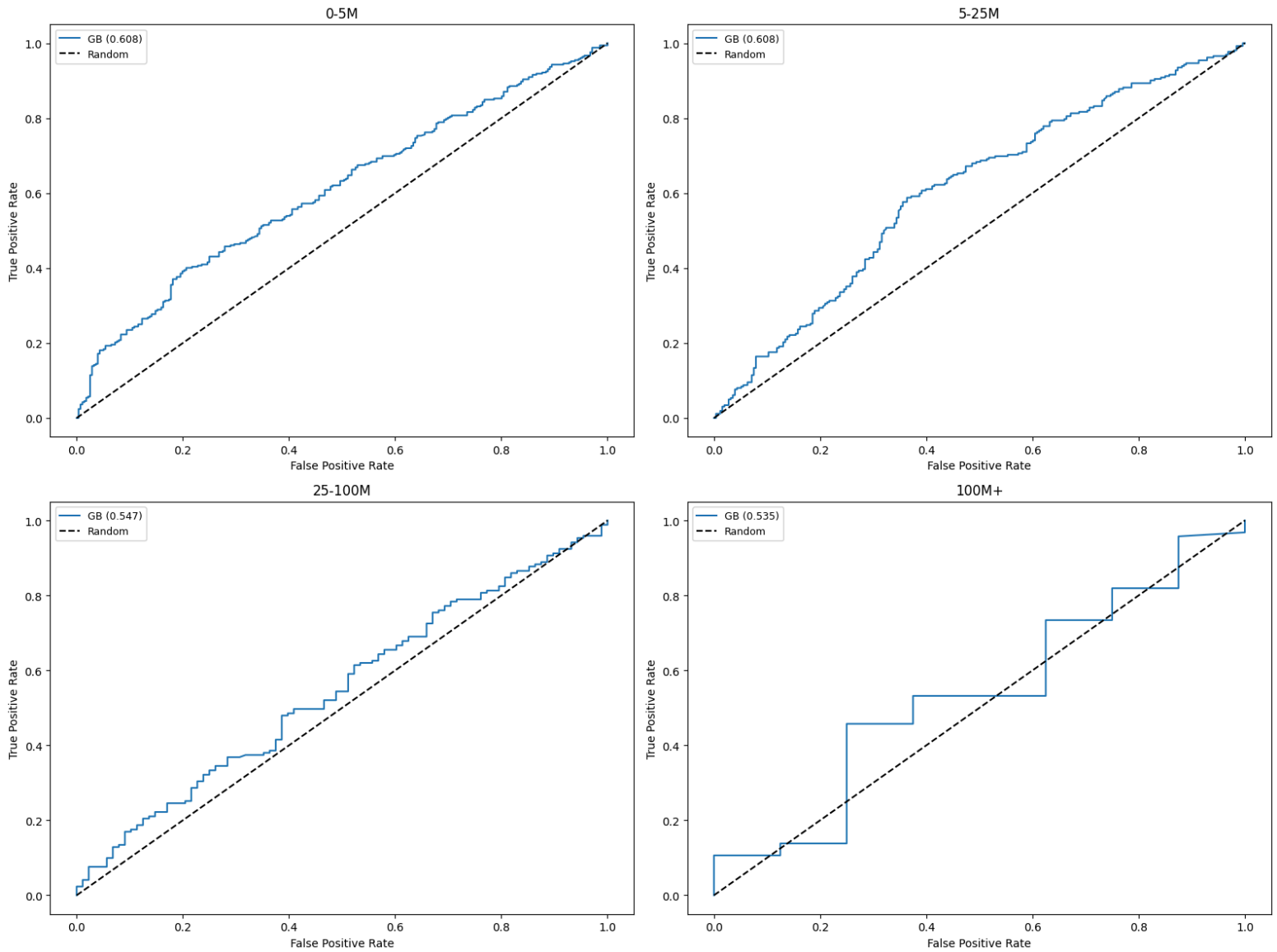
Overall Movie Success Rate broken out by Year, Budget Bucket, and Dataset



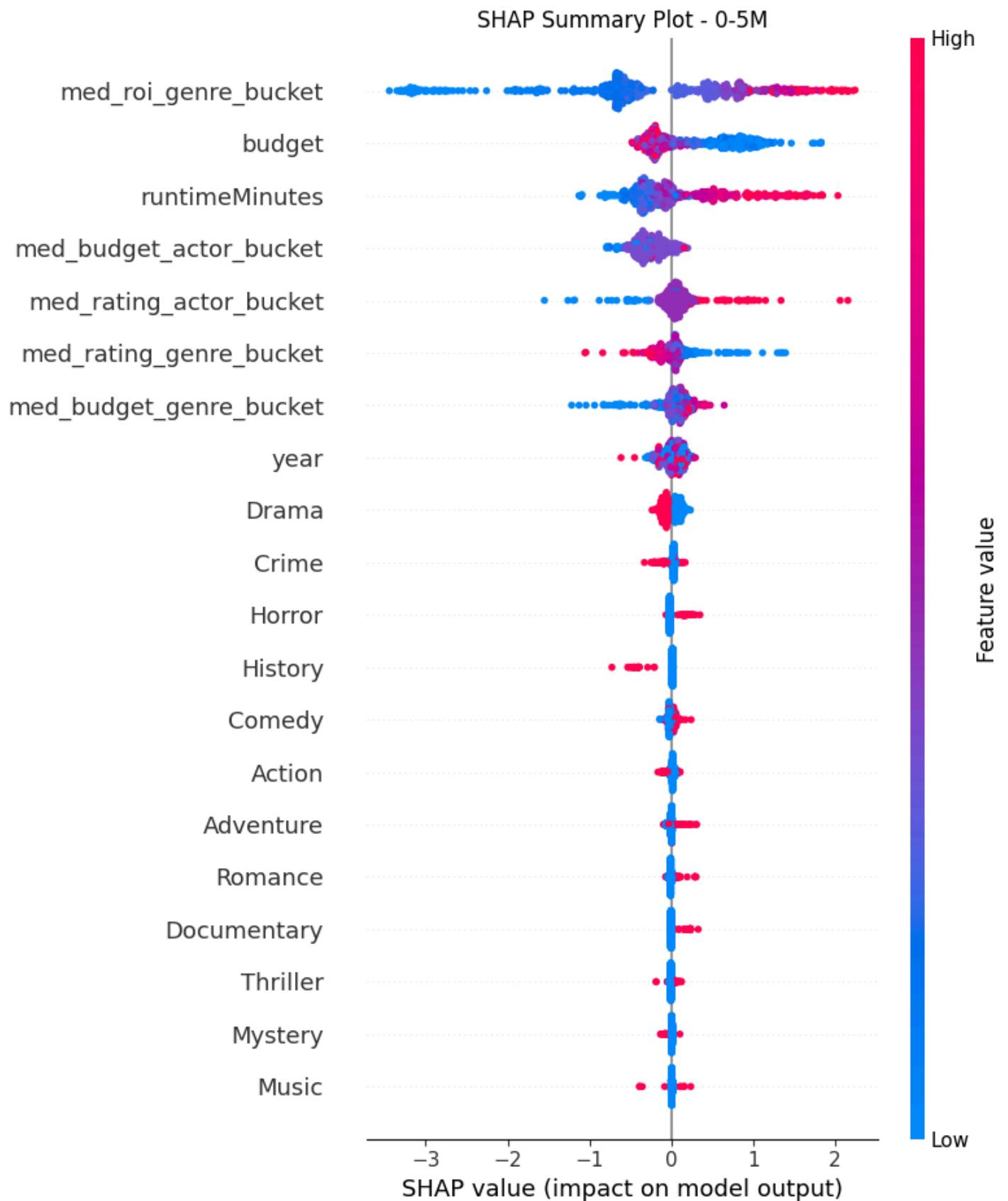
ROC Curves for Each Model and Budget Bucket on Validation Set



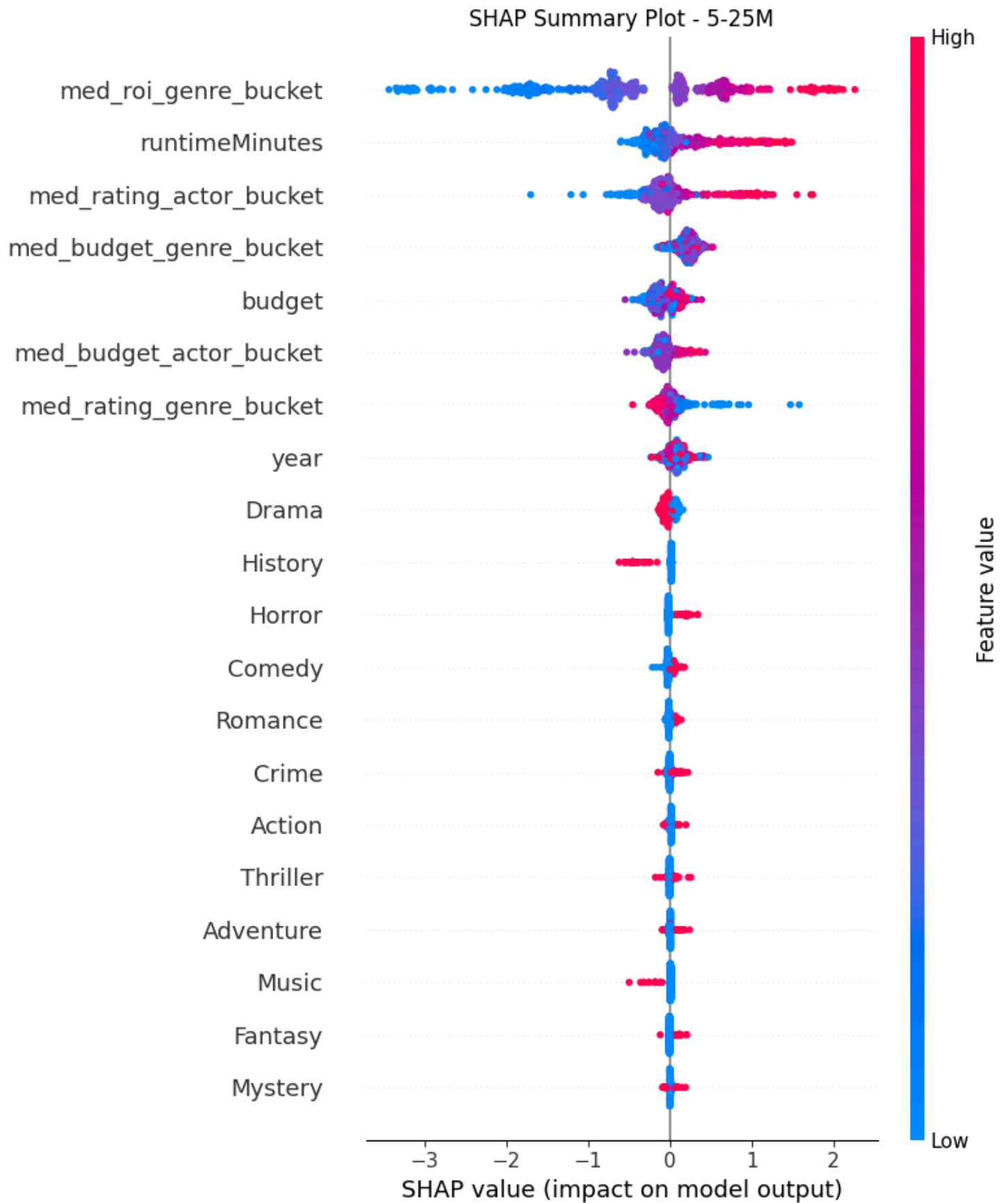
ROC Curves for Gradient Boosting Classifier (GBC) by Budget Bucket on Test Set



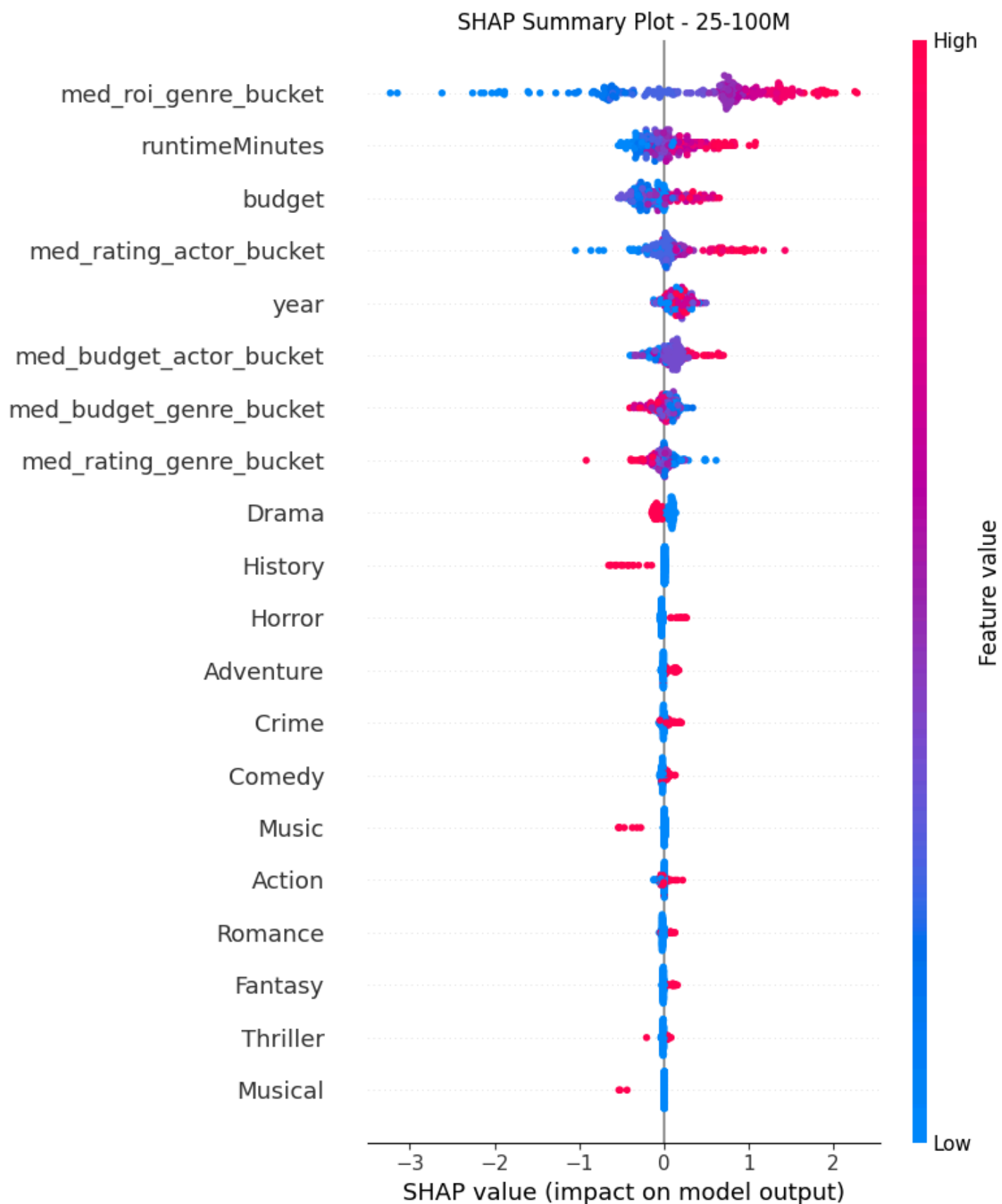
SHAP for GBC for \$0-5M Budget Segment on Test Set



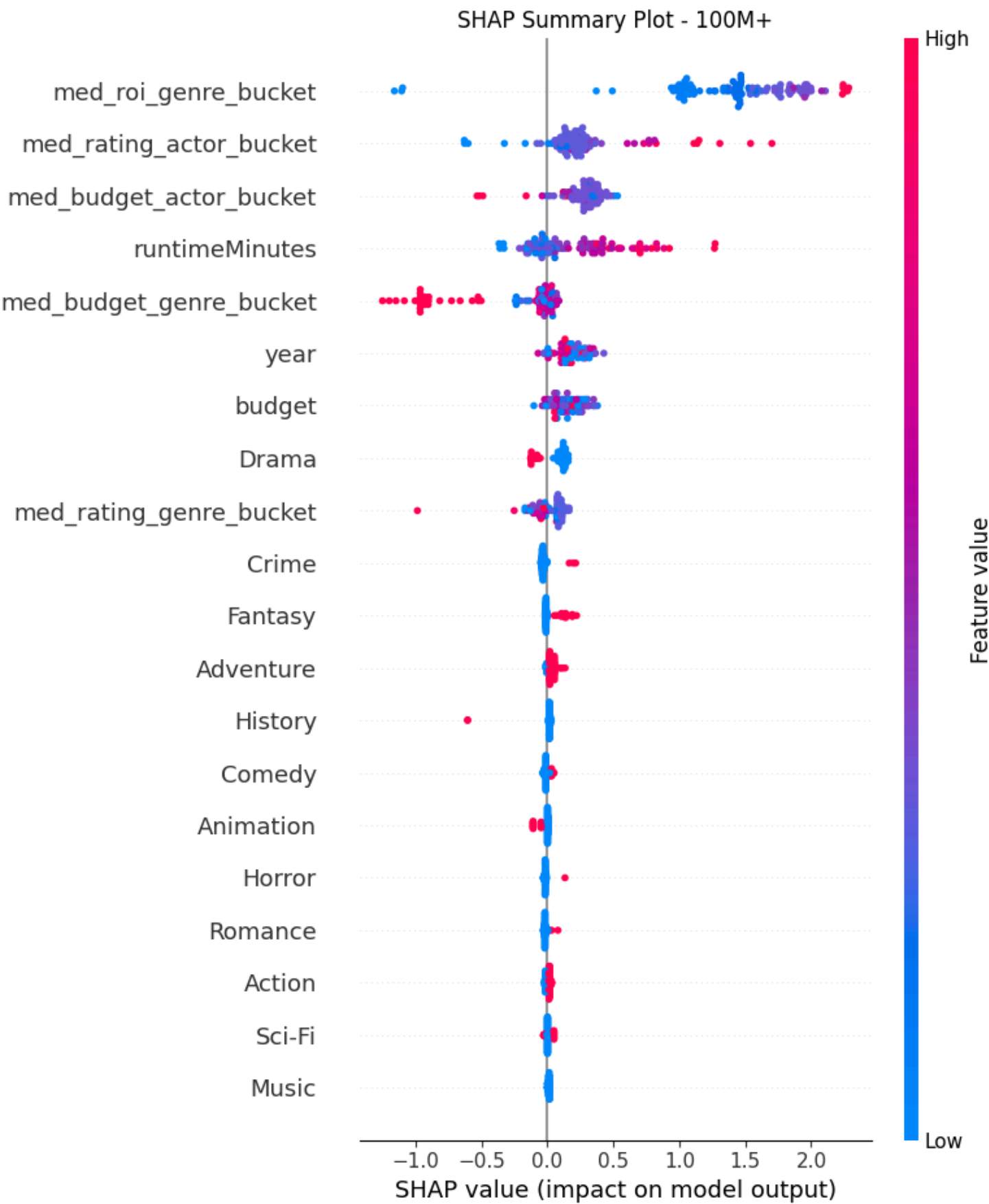
SHAP for GBC for \$5-25M Budget Segment on Test Set



SHAP for GBC for \$25-100M Budget Segment on Test Set



SHAP for GBC for \$100M+ Budget Segment on Test Set



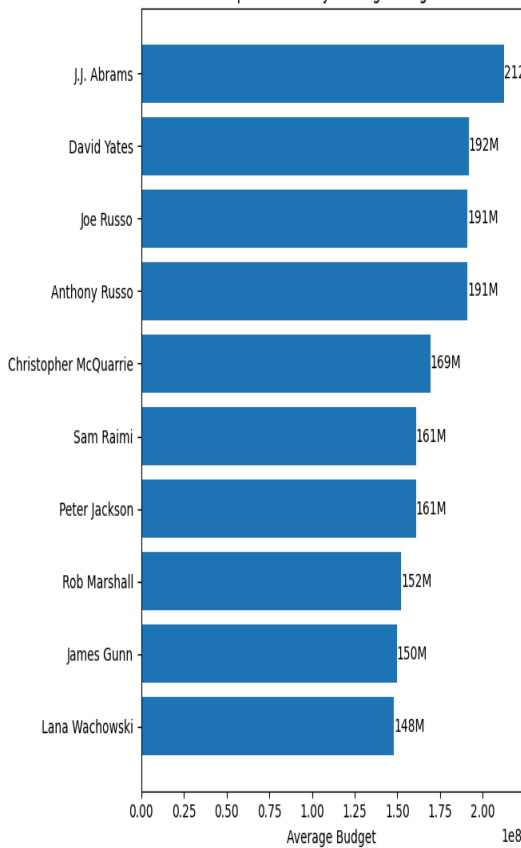
Director-Level Analysis

Directors exhibit a similar structural separation across performance dimensions. High-budget directors such as J. J. Abrams and David Yates are strongly associated with large-scale franchise productions, maintaining stable ROI (~3–4.5) and strong ratings (~7+). High-ROI directors such as James Wan operate within lower-budget, genre-driven films, achieving very high efficiency (ROI 8–10+). In contrast, top-rated directors such as Christopher Nolan and Quentin Tarantino produce critically acclaimed films with moderate ROI.

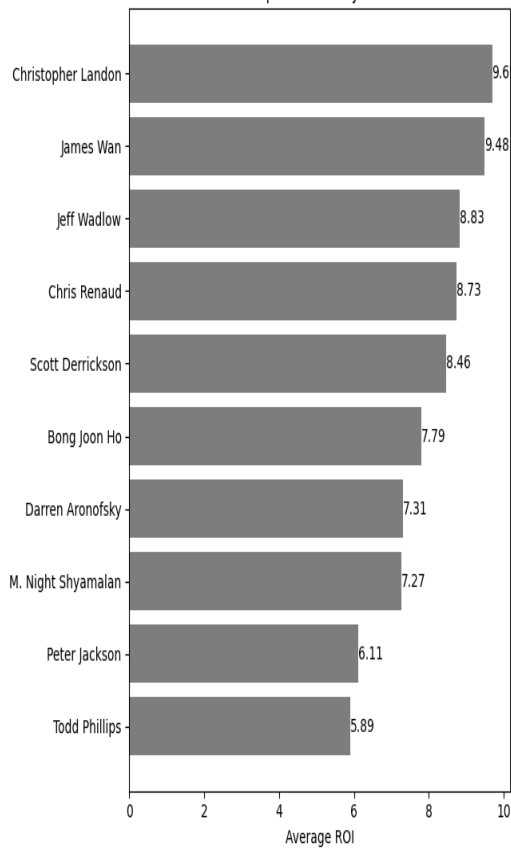
Over time, director performance follows a similar evolution as actors, moving from early alignment to increasing segmentation. Budget becomes more strongly tied to franchise ecosystems, ROI to cost-efficient genre strategies, and ratings to directorial vision and storytelling.

Director-level effects reinforce the need for aggregated historical features rather than single metrics. While directors contribute additional predictive signal, their impact is secondary to structural drivers such as budget and genre.

Top Directors by Average Budget



Top Directors by ROI



Top Directors by Rating

